

Safety data sheet

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BASF Safety data sheet

Date / Revised: 04.08.2025

Product: **Loxanol® MI 6311**

Version: 1.0

(30756558/SDS_GEN_SG/EN)

Date of print: 16.11.2025

1. Substance/preparation and manufacturer/supplier identification

Product name:

Loxanol® MI 6311

Use: additives for inks, varnishes or coatings, stabilizer, Paint additive

Recommended use: Chemical, for industrial use only

Manufacturer/supplier:

BASF South East Asia Pte Ltd.

128 Beach Road #18-01

Guoco Midtown, 189773, Singapore

Telephone: +65 8322 4420

Telefax number: +65 6 334-0330

E-mail address: benny.zou@basf.com

Emergency information:

Singapore Emergency Toll-Free Number:

Telephone: 1800-723-1361

International emergency number:

Telephone: +49 180 2273-112

2. Hazard identification

Classification of the substance and mixture:

No need for classification according to GHS criteria for this product.

Label elements and precautionary statement:

The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards which do not result in classification:

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

imidazole derivative, in water

No particular hazards known.

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:

Rinse mouth and then drink 200-300 ml of water.

Note to physician:

Symptoms: (Further) symptoms and / or effects are not known so far

Hazards: (Further) symptoms and / or effects are not known so far

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Suitable extinguishing media:

water spray, foam, dry powder, carbon dioxide

Specific hazards:

nitrogen oxides, carbon oxides

The substances/groups of substances mentioned can be released in case of fire. Under certain conditions in case of fire other hazardous combustion products may be generated.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions:

Handle in accordance with good industrial hygiene and safety practice.

Environmental precautions:

Discharge into the environment must be avoided.

Methods for cleaning up or taking up:

For small amounts: Pick up with suitable appliance and dispose of.

For large amounts: Pick up with suitable appliance and dispose of.

Incinerate or take to a special waste disposal site in accordance with local authority regulations.

7. Handling and Storage

Handling

Ensure thorough ventilation of stores and work areas. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Keep away from sources of ignition - No smoking.

Storage

Segregate from acids and acid forming substances.

Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

Storage stability:

Storage temperature: $\leq 40\text{ }^{\circ}\text{C}$

Storage duration: 12 Months

From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

8. Exposure controls and personal protection

Components with occupational exposure limits

No substance specific occupational exposure limits known.

Personal protective equipment

Respiratory protection:

Respiratory protection required in case of exceptional circumstances (e.g.: accidental release, exceeding the occupational exposure limit) Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
 butyl rubber (butyl) - 0.7 mm coating thickness

Manufacturer's directions for use should be observed because of great diversity of types.

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:

In order to satisfy general industrial hygiene rules safety glasses with side-shields (e.g. EN 166) are recommended.

Body protection:

No body protection required if used for intended purpose and satisfying generally accepted industrial hygiene rules.

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. Store work clothing separately. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to yellowish	
Odour:	amine-like	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
pH value:	9 - 11 (100 g/l, 20 °C) Information based on the main component/s.	
Melting point:	< -30 °C	
decomposition point:	> 150 °C (996 hPa) Information based on the main component/s., Product turns dark. The substance / product decomposes.	(OECD Guideline 103)
Flash point:	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2719, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 13736, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2592, open cup)

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Evaporation rate:	No data available.	
Flammability (solid/gas):	No data available.	
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	No data available.	
Information on: 1-(2-hydroxyethyl)imidazolidin-2-one		
Ignition temperature:	375 °C	(DIN 51794)

Thermal decomposition:	No data available.	
Information on: 1-(2-hydroxyethyl)imidazolidin-2-one		
Thermal decomposition:	380 °C , 140 kJ/kg	(DSC (OECD 113))

SADT:	Study scientifically not justified.	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Vapour pressure:	0.000013 hPa (25 °C) Information based on the main component/s.	(OECD Guideline 104)
Density:	1.311 g/cm ³ (20 °C, 1,013 hPa) Information based on the main component/s.	(OECD Guideline 109)
Relative density:	No data available.	
Relative vapour density (air):	Water content greater than 10%.	
Solubility in water:	Information based on the main component/s. 982 g/l (20 °C, 1,013 hPa)	
Miscibility with water:	completely (e.g. >=90%)	
Hygroscopy:	Non-hygroscopic	
Partitioning coefficient n-octanol/water (log Pow):	-0.7 (20 °C; pH value: 10) Information based on the main component/s.	(OECD Guideline 107)
Information on: 1-(2-hydroxyethyl)imidazolidin-2-one		
Adsorption/water - soil:	log KOC: -0.096	(calculated)

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Viscosity, dynamic: 27.4 mPa.s (ASTM D 7042)
(25 °C)
Viscosity, kinematic:
No data available.

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

Specific Surface Area:
No data available.

Particle Shape:
No data available.

Dustiness:
No data available.

10. Stability and Reactivity

Conditions to avoid:
Avoid extreme heat.

Thermal decomposition: No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Thermal decomposition: 380 °C, 140 kJ/kg (DSC (OECD 113))

Substances to avoid:
strong bases, strong acids

Corrosion to metals: Corrosive effects to metal are not anticipated.

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:
(oral):No data available.

Acute inhalation toxicity

(by inhalation):No data available.

Acute dermal toxicity

(dermal):No data available.

Assessment of acute toxicity

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Acute oral toxicity

Experimental/calculated data:

LD50 rat (oral): > 2,000 mg/kg (similar to OECD guideline 401)

Limit concentration test only (LIMIT test). No mortality was observed.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Acute dermal toxicity

Experimental/calculated data:

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)

Limit concentration test only (LIMIT test). No mortality was observed.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of acute toxicity

Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact.

Symptoms

(Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

No data available.

Experimental/calculated data:

Skin corrosion/irritation:No data available.

Serious eye damage/irritation:No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of irritating effects:

Not irritating to the skin. Not irritating to the eyes.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Experimental/calculated data:

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

No data available.

Experimental/calculated data:

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:

Based on available data, the classification criteria are not met.

Experimental/calculated data:

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of carcinogenicity:

No data was available concerning carcinogenic activity.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available data, the classification criteria are not met.

Experimental/calculated data:

No data available.

Developmental toxicity

Assessment of teratogenicity:

Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on available data, the classification criteria are not met.

Experimental/calculated data:

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment of repeated dose toxicity:

Repeated oral uptake of the substance did not cause substance-related effects.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

No data available.

Aquatic invertebrates:

No data available.

Aquatic plants:

No data available.

Microorganisms/Effect on activated sludge:

No data available.

Chronic toxicity to fish:

No data available.

Chronic toxicity to aquatic invertebrates:

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Toxicity to fish:

LC50 (96 h) 10,250 mg/l, *Brachydanio rerio* (OECD 203; ISO 7346; 84/449/EWG, C.1, static)

The statement of the toxic effect relates to the analytically determined concentration.

LC50 (96 h) > 1,004 mg/l, *Oncorhynchus mykiss* (OECD Guideline 203, semistatic)

The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Aquatic invertebrates:

EC50 (48 h) > 100 mg/l, *Daphnia magna* (Directive 79/831/EEC, static)

The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Aquatic plants:

EC50 (72 h) > 1,059 mg/l (growth rate), *Pseudokirchneriella subcapitata* (OECD Guideline 201, static)

The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Microorganisms/Effect on activated sludge:

EC20 (30 min) > 1,000 mg/l, activated sludge, domestic (DIN EN ISO 8192)

Nominal concentration.

Mobility

Assessment transport between environmental compartments:

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment transport between environmental compartments:

The substance will not evaporate into the atmosphere from the water surface.

Adsorption to solid soil phase is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Assessment biodegradation and elimination (H₂O):

Not readily biodegradable (by OECD criteria). Inherently biodegradable. The product is biodegradable after extended adaptation.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one

Elimination information:

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< 10 % DOC reduction (28 d) (DIN EN ISO 7827) (municipal sewage treatment plant effluent)

> 90 % DOC reduction (28 d) (OECD Guideline 302 B) (activated sludge, domestic)

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one
Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one
Information on Stability in Water (Hydrolysis):
 $t_{1/2} > 365$ d (pH 7, 25 °C, pH value 7), (calculated)

Bioaccumulation potential

Assessment bioaccumulation potential:
No data available.

Information on: 1-(2-hydroxyethyl)imidazolidin-2-one
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Additional information

Other ecotoxicological advice:
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The local regulations on waste-water treatment must be followed.

13. Disposal Considerations

Incinerate in suitable incineration plant, observing local authority regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Domestic transport:

UN number or ID number	Not classified as a dangerous good under transport regulations
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Other regulations**

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.